

ADVANCED PROGRAMMING PROJECT - TEST PLAN

ANON

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1. INTRODUCTION

This test plan details the methods and examples I have used to verify whether the methods from my project are functioning to the specifications in the project description. The methods tested are multiply, inverse and divide as required.

2. BICOMPLEX NUMBER

This class is designed to create bicomplex numbers from 4 real numbers or 2 complex numbers. It is also designed to operate on these bicomplex numbers.

Where non-trivial, the expected outputs were calculated by hand.

2.1. `BicomplexNumber.multiply(BicomplexNumber w)`. This function returns the product of two bicomplex numbers, z and w . Generic examples of two bicomplex numbers are tested including negative and decimal. A zero bicomplex number is also tested.

z	w	Expected output
$1 + 1i + 1j + 1k$	$1 + 1i + 1j + 1k$	$4k$
$1 + 1i + 1j + 1k$	$0 + 0i + 0j + 0k$	$0 + 0i + 0j + 0k$
$1 + 5i + 8j + 6k$	$9 + 2i + 7j + 9k$	$-3 - 67i + 22j + 114k$
$9 + 2i + 7j + 9k$	$1 + 5i + 8j + 6k$	$-3 - 67i + 22j + 114k$
1	$-4 + 0i + 7j - 1k$	$-4 + 0i + 7j - 1k$
$4 + 8i + 2j + 6k$	$0.25 + 2i + 0.5j + 4k$	$8 - 1i + 41.5j + 25.5k$

2.2. BicomplexNumber.invert(). This function returns the inverted form of a bicomplex number z.

Generic examples of bicomplex numbers are tested including negative and decimal.

Exceptional case where the bicomplex number is not invertible is also tested.

z	Expected output
$0 + 0i + 0j + 0k$	ArithmeticException: "Bicomplex number is not invertible!"
$1 + 1i + 1j + 1k$	$0.25 + -0.25i - 0.25j + 0.25k$
$0.25 + 0.25i + 0.25j + 0.25k$	$1 + 1i + 1j + 1k$
$2 - 2i - 2j + 2k$	$0.125 + 0.125i + 0.125j + 0.125k$

2.3. BicomplexNumber.divide(BicomplexNumber w). This function divides a bicomplex number z by another bicomplex number w.

Generic examples of bicomplex numbers are tested including negative.

Exceptional case where the bicomplex number w is not invertible is also tested.

z	w	Expected output
$0 + 0i + 0j + 0k$	$1 + 1i + 1j + 1k$	$0 + 0i + 0j + 0k$
$1 + 1i + 1j + 1k$	$0 + 0i + 0j + 0k$	ArithmeticException: "Bicomplex number is not invertible!"
$1 + 1i + 1j + 1k$	$1 + 1i + 1j + 1k$	1
$1 + 1i + 1j + 1k$	1	$1 + 1i + 1j + 1k$
$-3 - 67i + 22j + 114k$	$9 + 2i + 7j + 9k$	$1 + 5i + 8j + 6k$